

① Bitwise Algorithm :-

1. OR (1) :-

$$0 \mid 1 = 1$$

$$1 \mid 0 = 1$$

$$1 \mid 1 = 1$$

$$0 \mid 0 = 0$$

2. AND ($\&$) :-

$$0 \& 1 = 0$$

$$0 \& 0 = 0$$

$$1 \& 0 = 0$$

$$1 \& 1 = 1$$

3. XOR (n) :-

$a \wedge b \Rightarrow$

$$1 \wedge 1 = 0$$

$$0 \wedge 0 = 0$$

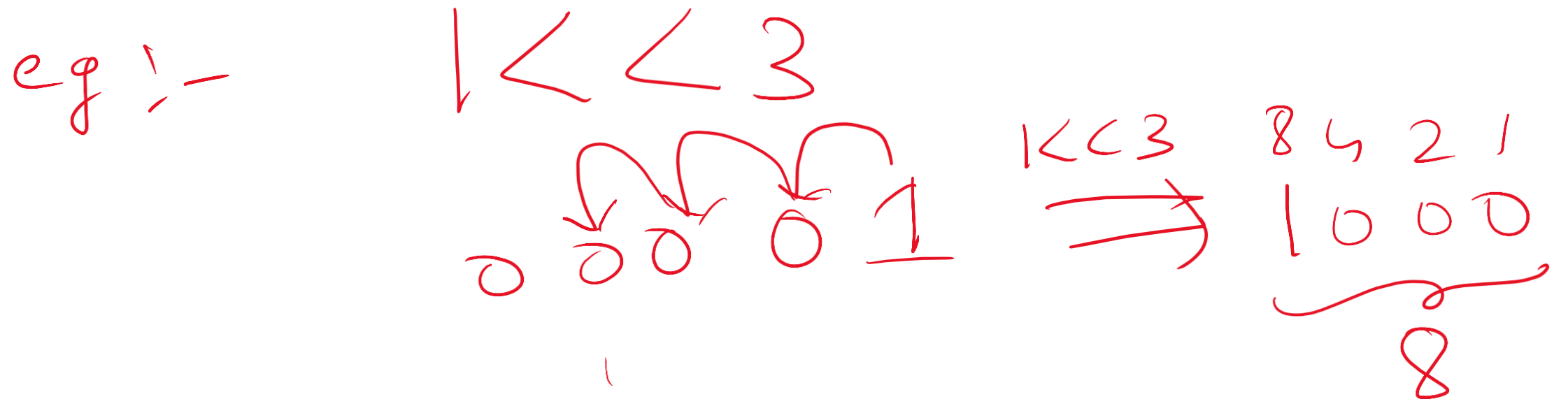
$$1 \wedge 0 = 1$$

$$0 \wedge 1 = 1$$

1 0	1 1
1 1	0 0
0 1	1 1

① In XOR operator, if both the bits are same, then in result that specific bit will be 0 otherwise it will be 1.

4. Left shift ($1 < n$) $\Rightarrow$ This is going to shift number 1 to the left by n bits.



5. Right Shift ($\gg n$) :- It implies that you are going to shift set bit by n places towards right.

$$8 = \underbrace{1000}_{1000} \xrightarrow{\gg 2} \underbrace{0010}_{0010}$$

6. Complement ($\sim$) :- $\sim 1 = 0$; $\sim 0 = 1$

○ Tricks :-

1. To set a bit at n^{th} position in the number, you can do it via

following ways :-

- (i) First left shift '1' to n^{th} position using $1 \ll n$.
- (ii) After this, use Bitwise OR bcoz you already shifted '1' to n^{th} position.

0

$$\underline{\text{num}} = \left(\underline{\text{num}} \mid (1 \ll h) \right)$$

2. To clear a bit at h^{th} position
in "num", use :-

①

$$\text{num} = \text{num} \& \left(\sim (1 \ll h) \right)$$

1
0

3. How to check if n^{th} bit is set or not :-

① $\text{num} \& (1 \ll n) \rightarrow 0$

19 =

&

16	8	4	2	1
1	0	1	1	1
0	0	1	0	0
0	0	1	0	0
4				

4. To make lowest set bit in a number,

$$n \& (n-1)$$

$$n = 10$$

$$n-1 = 9$$

msb ←					→	lsb	
16	8	4	2	1			
0	1	0	1	0	↗		
&	0	1	0	0	1		
					1 0 0 0		
						↖	

5. Finding lowest set bit of
a number n :-

$$\text{res} = n \& \neg n$$

① 2 's complement = 1 's complement + 1

① 2's complement is used to find binary representation of negative



~~W~~

$20 =$

1's complement =

+

-20 =

$\begin{array}{r} 10\downarrow00 \\ + 01011 \\ \hline 00001 \\ \hline 01100 \end{array}$

$\frac{n2-n}{20} = \downarrow 00100$

⑥ SNS =

$$\cancel{S \wedge S} \quad S = \bar{S}$$

A hand-drawn diagram illustrating a queue using an array. A horizontal line represents the array. Three circles are drawn inside, representing elements. A checkmark is drawn to the left of the first circle.

① $\begin{array}{cccccccccccc} 5 & 4 & 5 & 4 & 6 & 6 & 3 & 3 \\ \hline & & & & & & & \\ \hline & & 0 & & & & & \end{array}$

$$\textcircled{D} \quad arr = [\underline{5}, \underline{6}, \underline{5}, \underline{7}, \underline{9}, \underline{6}]$$

$$ans = [\underline{7}, \underline{9}]$$

$$\underbrace{xor}_{xor} = 5 \wedge 7 =$$

$$a = a \wedge arr(i);$$

$$b = b \wedge arr(i);$$

$$\begin{array}{r}
 \begin{array}{cccc}
 & 8 & 4 & 2 & 1 \\
 & & 1 & 0 & 1 \\
 \wedge & & 1 & 1 & 1 \\
 \hline
 & & 0 & 1 & 0 \\
 \hline
 & & & \uparrow &
 \end{array}
 \end{array}$$

① arr = [14, 5, 14, 14]

ans = 5 for (int i = 0; i <= 3; i++) ③

{ x = 1 < i }

10 9 8 7 6 5 4 3 ② 1 0

Bit No = 31 30

for (int j = 0; j <= i; j++)

{ if (arr[j] & x) 14 =

sum++;

}

14 = 5 =

if (sum % 3 == 0)

ans = ans * x;

311 for i, else

1	1	1	0
1	1	1	0
1	1	1	0
0	1	0	1
3	4	3	1